RvS: What is Essential for Offline RL via Supervised Learning?

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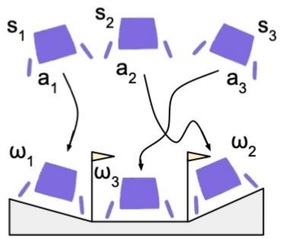
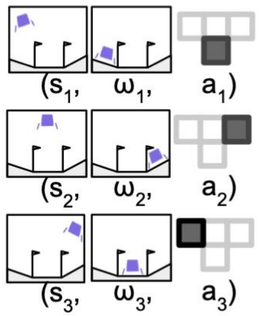
BERKELEY ARTIFICIAL INTELLIGENCE RESEARCH

We can Reduce RL to Supervised Learning! [1-3]



Step 1 Collect (offline) replay buffer of states *s*, actions *a*, and outcomes ω

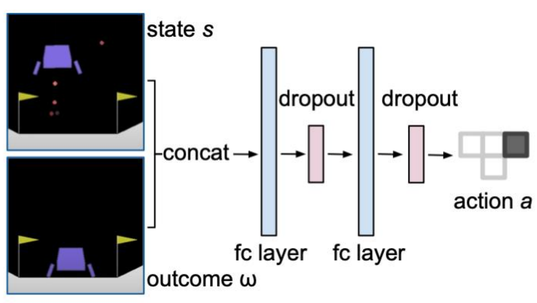


Step 2 Perform hindsight relabeling

For all trajectories: For all timesteps in that trajectory: For all achieved outcomes:

$$\max_{\theta} \sum_{\tau \in \mathcal{D}} \sum_{1 \le t \le |\tau|} \mathbb{E}_{\omega \sim f(\omega \mid \tau_{t:H})}[\log \pi_\theta(a_t \mid s_t, \omega)]$$

Step 3 Maximize likelihood of the actions, given state and outcome



Supervised Learning has Many Possible Benefits

- More stable than RL
- (Comparatively) easy to debug and validate
- Success learning from large, precollected datasets

Conundrum: What's Important to Make It Work?

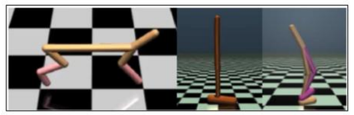
- Carefully reweighting the training data (RCP: Kumar *et al.*, 2019)
- Iterative, online data collection (GCSL: Ghosh *et al.*, 2021)
- Transformer sequence models (DT: Chen et al., 2021)

Key Questions of Our Work

1. Which design decisions are critical for RL via supervised learning?
2. How well does RL via supervised learning actually work?
3. What should we condition on? Does it matter?

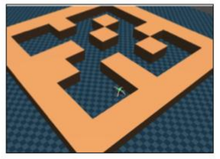
Tasks and Datasets

D4RL Gym Locomotion v2
Challenge: learn from (mixture) datasets from policies of varying quality.

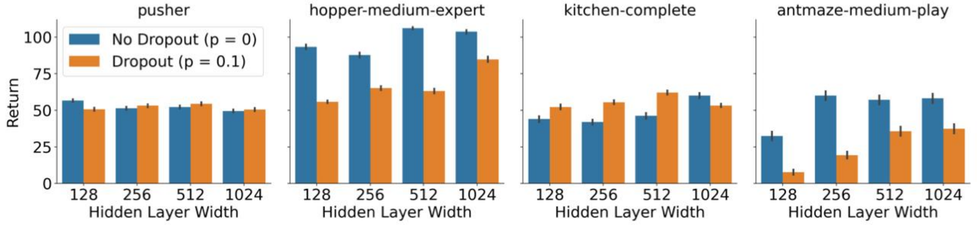


D4RL Frank Kitchen v0
Challenge: learn from human demonstrations that don't solve all subtasks in sequence.

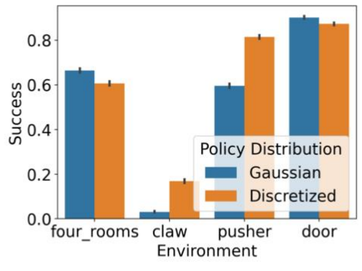
AntMaze v2
Challenge: stitch together parts of different trajectories.



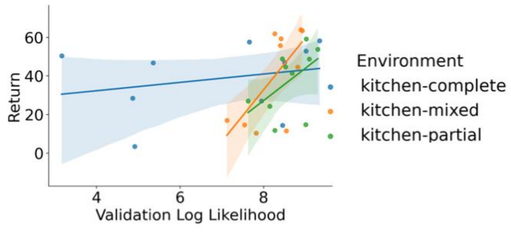
Experiments



Large network capacity, and sometimes dropout, are key for good performance.



Discretizing the action space generally helps.

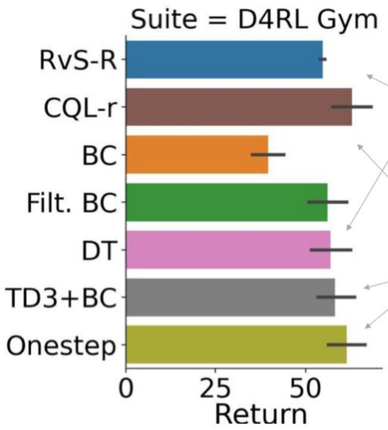


Validation loss is only loosely correlated with final performance.

Important Ingredients we Identify

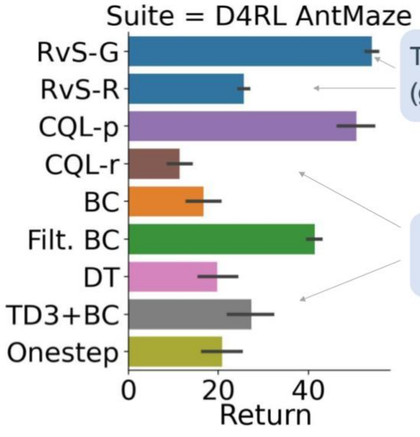
- ✓ Capacity & regularization
- ✓ Policy distribution
- ✓ Conditioning variable
- X Reweighting training data
- X Iterative, online data collection
- X Transformer sequence models

Overall Performance



RvS using just an MLP matches Decision Transformer!

TD learning methods perform especially well on random data.



The choice of conditioning variable (goal states vs rewards) is key.

RvS-G matches the dynamic programming methods!

References

1. Aviral Kumar, Xue Bin Peng, and Sergey Levine (2019). "Reward-Conditioned Policies." arXiv:1912.13465.

2. Dibya Ghosh, Abhishek Gupta, Ashwin Reddy, Justin Fu, Coline Manon Devin, Benjamin Eysenbach, and Sergey Levine (2021). "Learning to Reach Goals via Iterated Supervised Learning." In: ICML.

3. Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Michael Laskin, Pieter Abbeel, Aravind Srinivas, and Igor Mordatch (2021). In: NeurIPS.

Paper & Code: